F2 AI 工作坊 2 / Workshop 2

Python × Colab：數據與線性回歸
Python × Colab: data and linear regression

探索數據、讀模型表、比較 R^2

Explore data, read a model table and compare R^2

Python 和 Jupyter Notebook
Python and Jupyter Notebook

Python：用程式給電腦指令，例如「畫出這些點」

Python: instructions such as “plot these dots”

Jupyter Notebook：把說明、程式格和結果放在一起

Jupyter Notebook: explanations, code cells and results together

今天只需按播放、看結果；不用背誦程式

Today: press play and inspect the result. No memorising code

程式在哪裏跑？雲端運算
Where does code run? Cloud computing

雲端運算：透過網絡，使用遠端電腦的運算能力

Cloud computing: use a remote computer over the internet

Colab：在瀏覽器開啟 Notebook，讓雲端電腦執行程式

Colab: open a notebook in a browser; run code on a cloud computer

你的畫面顯示指令和結果

Your screen shows instructions and results

借用工具箱：scikit-learn
Borrow a toolbox: scikit-learn

函式庫：別人準備好的工具，讓我們重用

A library is a collection of tools we can reuse

scikit-learn（程式中叫 sklearn）提供 LinearRegression

scikit-learn, imported as sklearn, provides LinearRegression

fit：從數據找直線；predict：用直線作估計

fit: find the line. predict: use it for an estimate

開啟你的數據實驗簿
Open your data notebook

開啟老師的 Colab 連結，儲存自己的副本

Open the teacher’s Colab link and save your own copy

三節、五個程式格：探索 → 建模 → A、B、C 比較

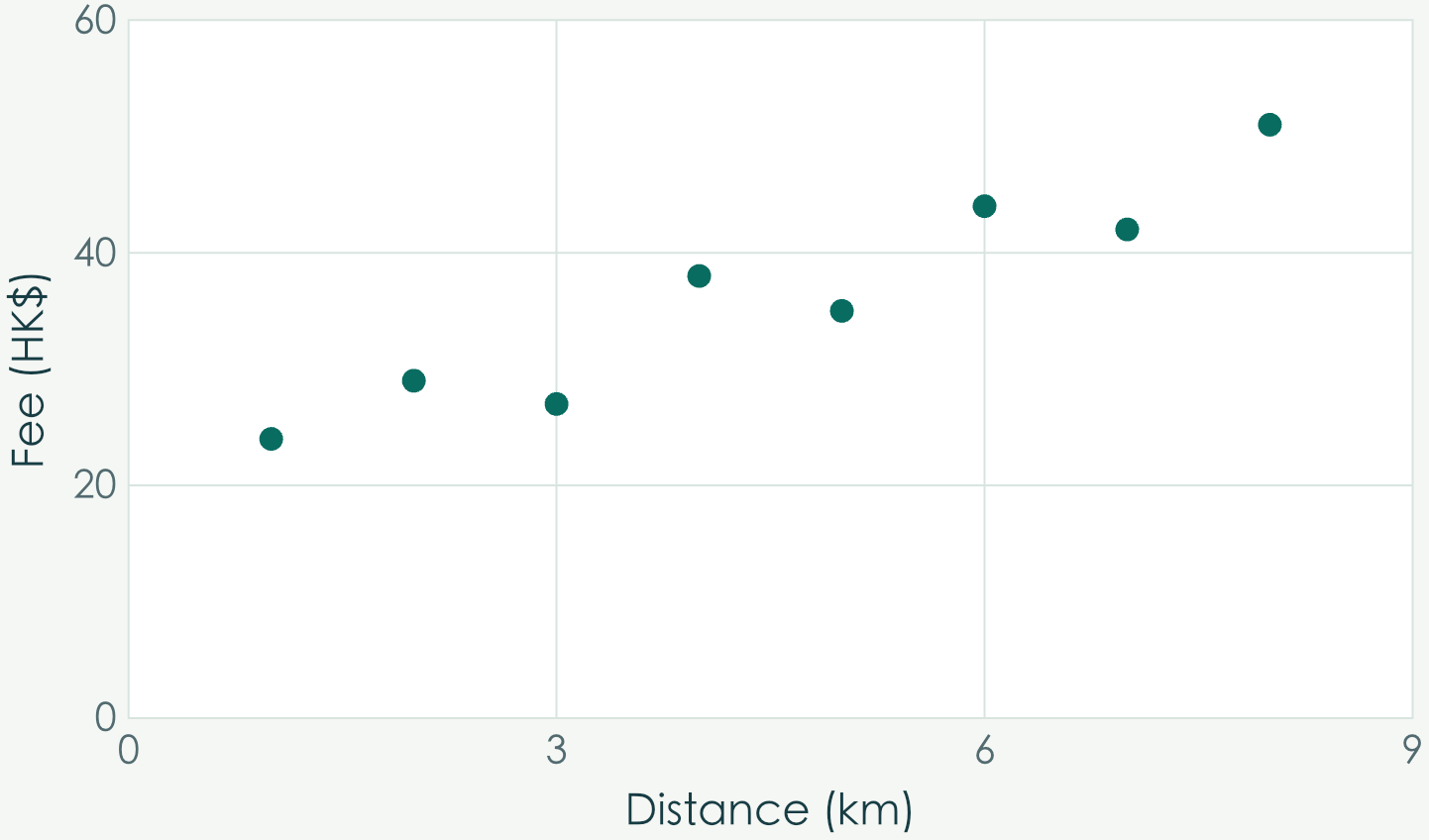
Three sections, five code cells: explore → fit → compare A, B, C

按每格左邊的 ▶，等待結果；工作紙只記結果和觀察

Press ▶️ and wait. Record results and observations on the worksheet

Colab: tinyurl.com/26jdbxfb

第 1 節：讓散點說話
Section 1: let the dots tell their story



整體關係是怎樣？

What is the overall relationship?

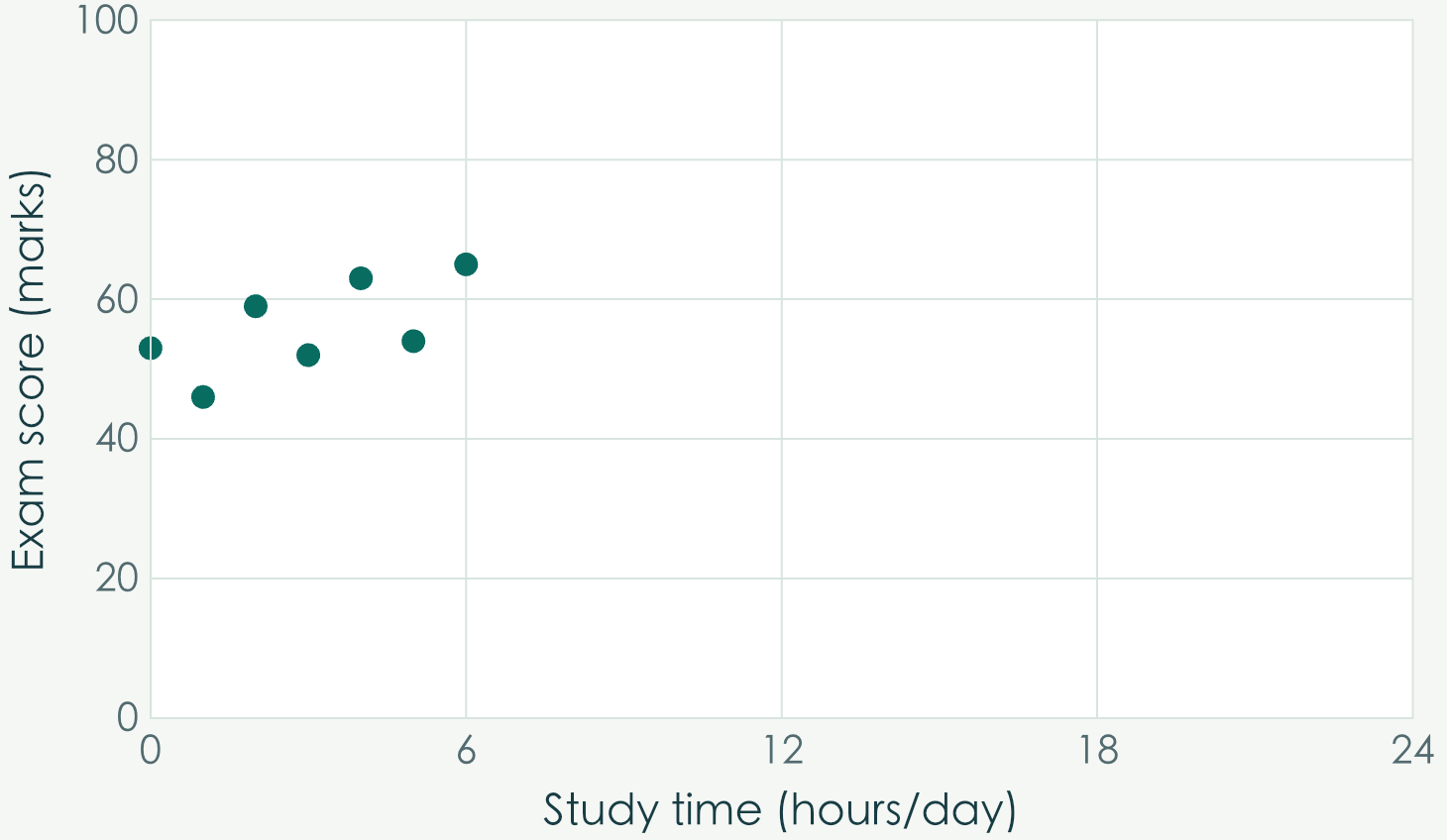
有例外嗎？找兩筆紀錄

Any exceptions? Use two records

將觀察記在工作紙

Record your observations

第 2 節：換一組學生數據
Section 2: try a different dataset



每點是一位模擬學生

Each dot is a simulated student

點是否排成完美直線？

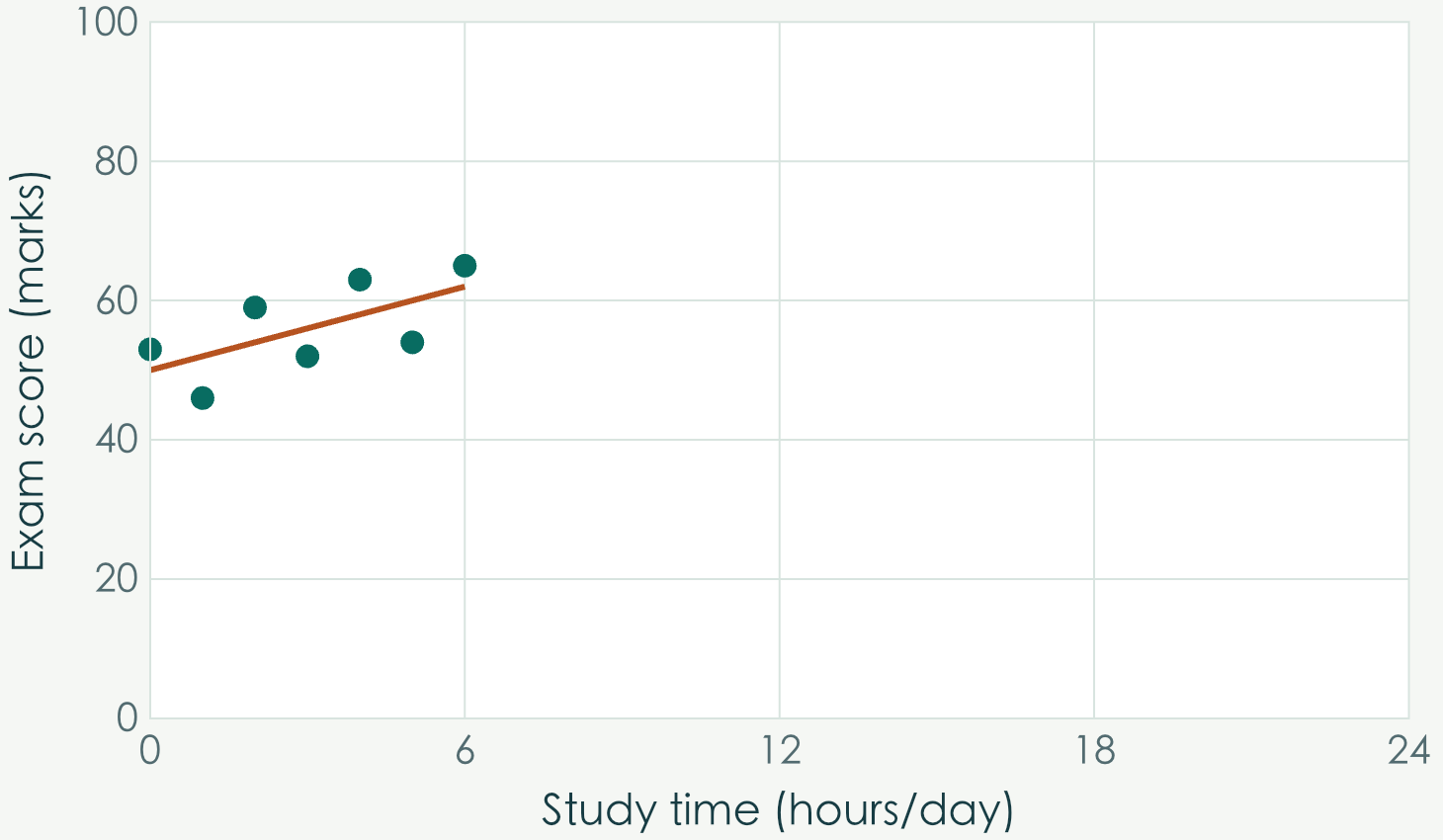
Do the dots form a perfect line?

用一條線概括整體趨勢

Use a line to summarise the trend

一條直線，概括整體趨勢

One line summarises the overall trend



y = mx + c

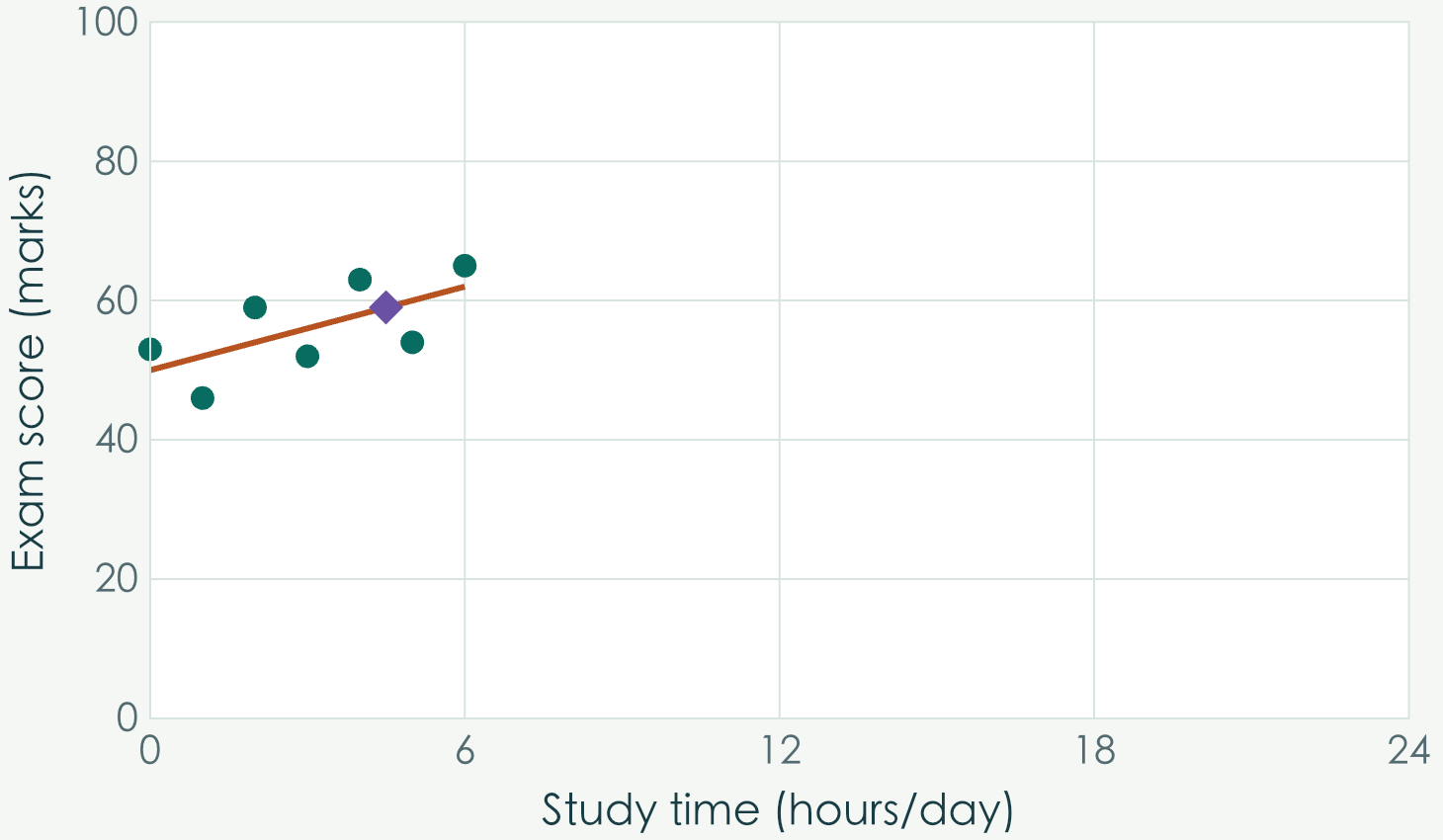
m：每多一小時的估計改變

m: estimated change per extra hour

c：0 小時的估計分數

c: estimated score at 0 hours

第 2 節：讀模型表，再找預測點
Section 2: read the table and prediction



執行第 2 節，抄下模型表

Run Section 2 and copy the table

紫色點：每天 4.5 小時

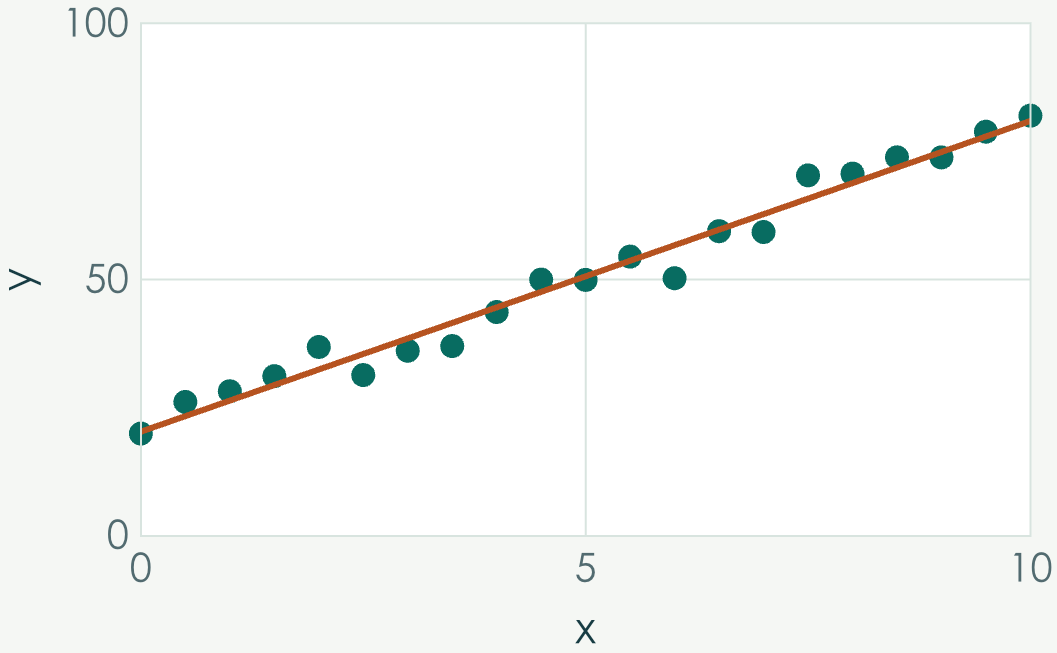
Purple dot: 4.5 hours per day

預測是估計，不是保證

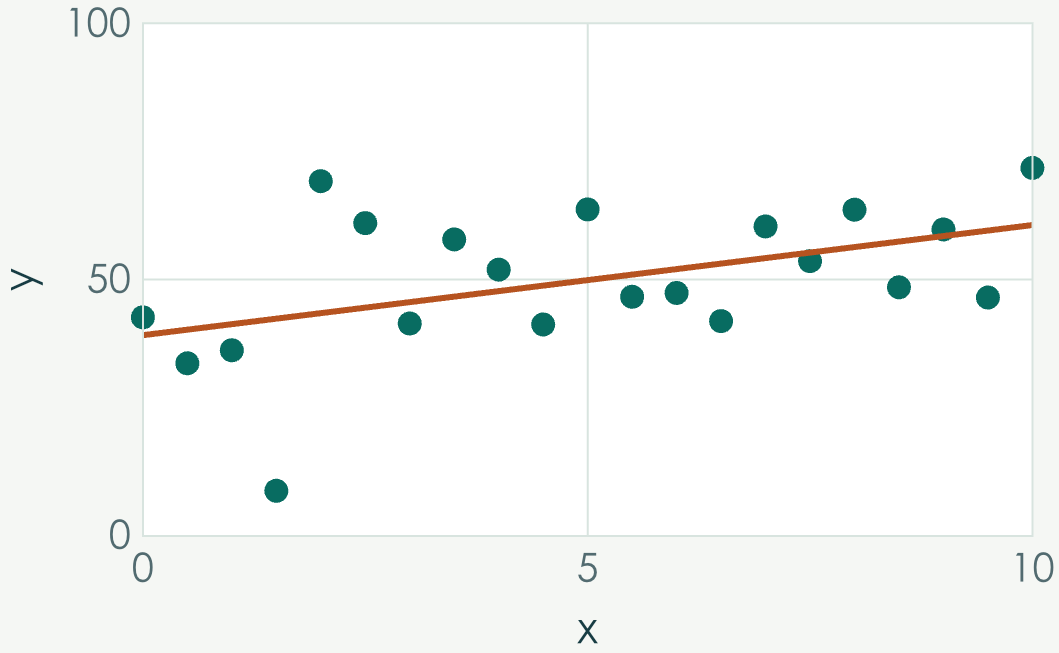
An estimate, not a guarantee

哪條線更能概括自己的數據？
Which line better summarises its data?

Example B



Example A



看看散點和直線 / Compare the dots with the line

R^2：給「擬合程度」一個數字
R^2: a number describing the fit

差異：各個 y 值有高有低，例如學生的成績不同

Variation: the y-values differ, such as students' scores

0 **R^2** **1**

接近 0：解釋較少差異

Near 0: little variation explained

接近 1：解釋較多差異

Near 1: more variation explained

在本課，R^2 是在用來建立模型的數據上計算

Here, R^2 is measured on the data used to fit the line

模型表：今天聚焦 R^2
The model table: focus on R^2

Metric	Value
n	7
m	2.00
c	50.00
R^2	**0.418**

教學模擬數據 / Simulated teaching data

直線解釋了約 42% 的成績差異

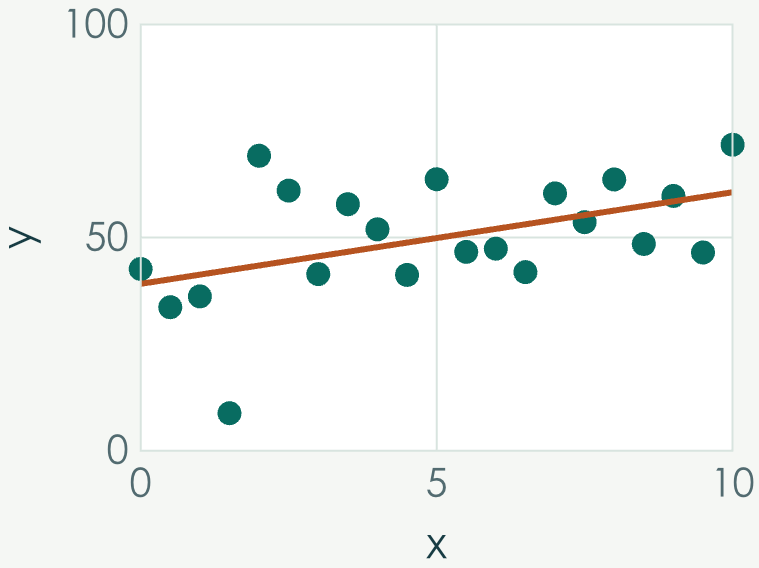
About 42% of score variation is explained

並非 42% 的預測答對

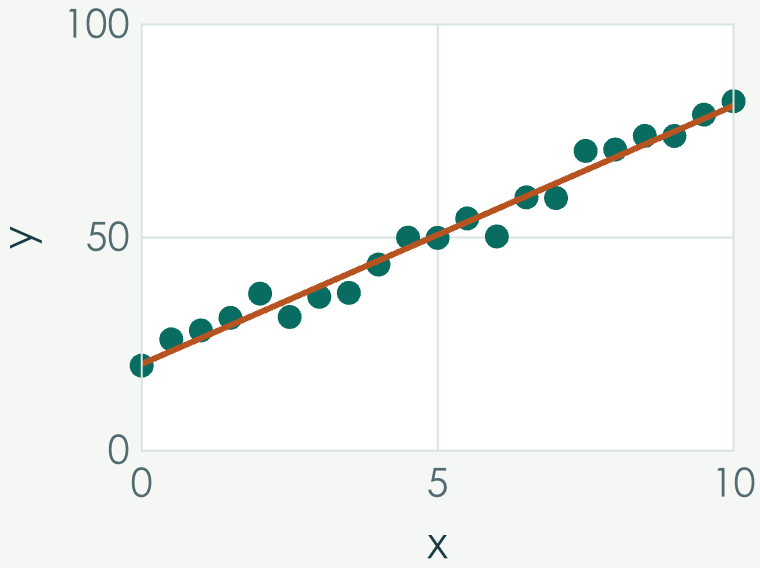
It does not mean 42% of predictions are correct

第 3 節：比較 A、B、C
Section 3: compare A, B and C

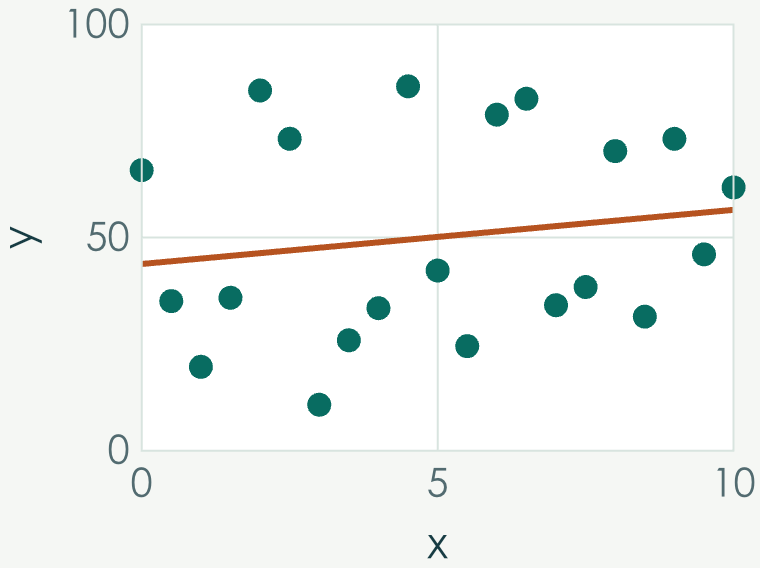
Example A



Example B



Example C

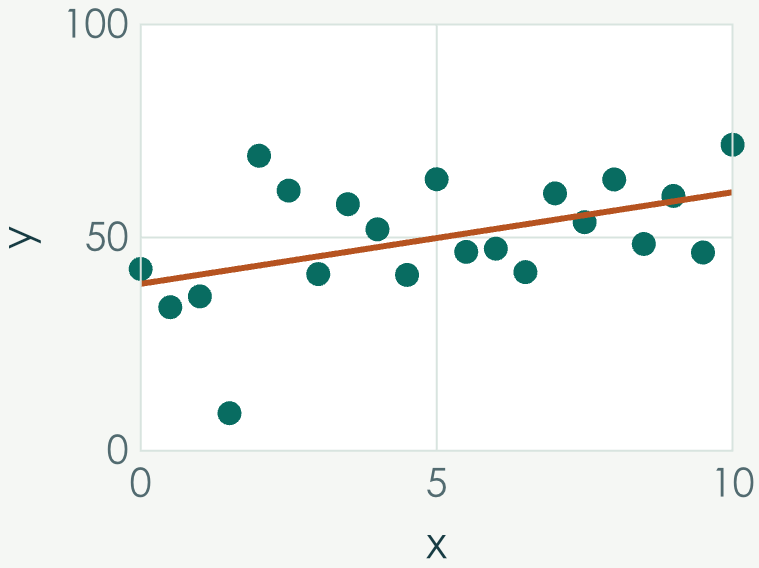


依次執行三格，記下 R^2，再排名 1、2、3

Run the three cells, record R^2, then rank 1, 2 and 3

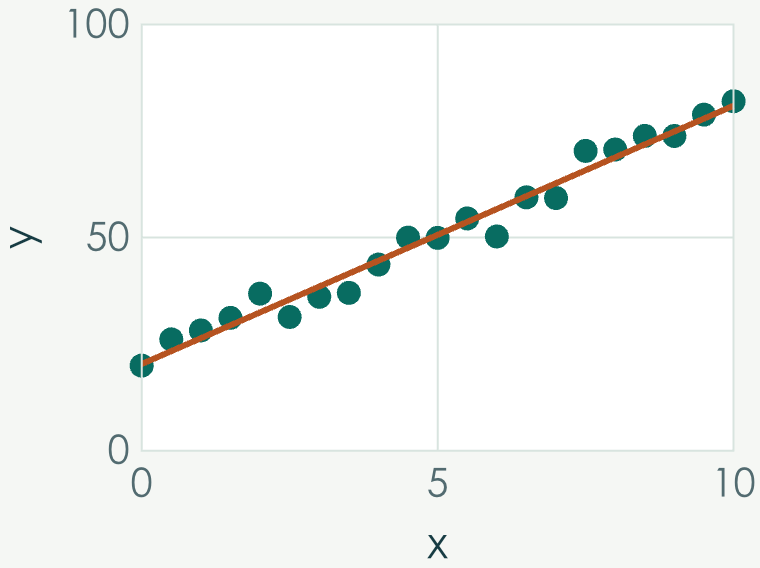
揭曉：比較解釋數據的比例
Reveal: compare the variation explained

Example A



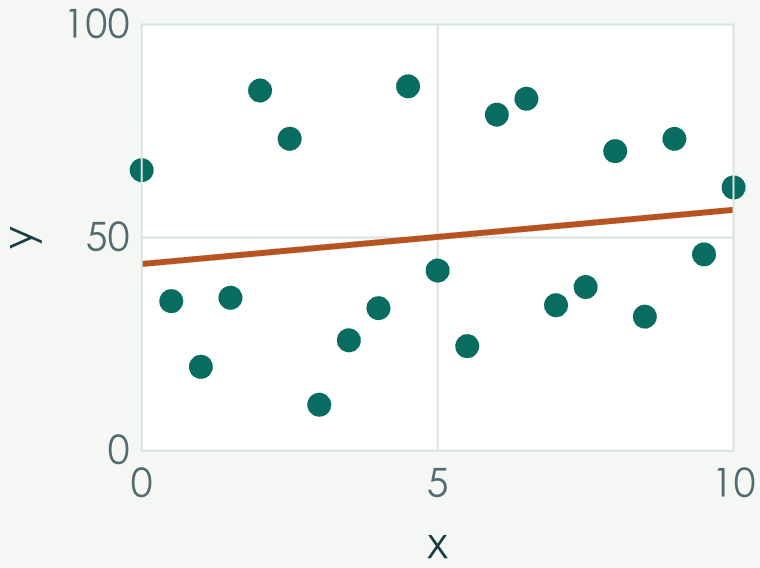
$R^2 = 0.218$

Example B



$R^2 = 0.977$

Example C



$R^2 = 0.027$

B → A → C　較強 / Stronger ·　較弱 / Weaker ·　隨機 / Random

R^2 很高，就一定猜得準嗎？
Does high R^2 guarantee a good prediction?

較高 R^2：這條線對建模數據的擬合較好

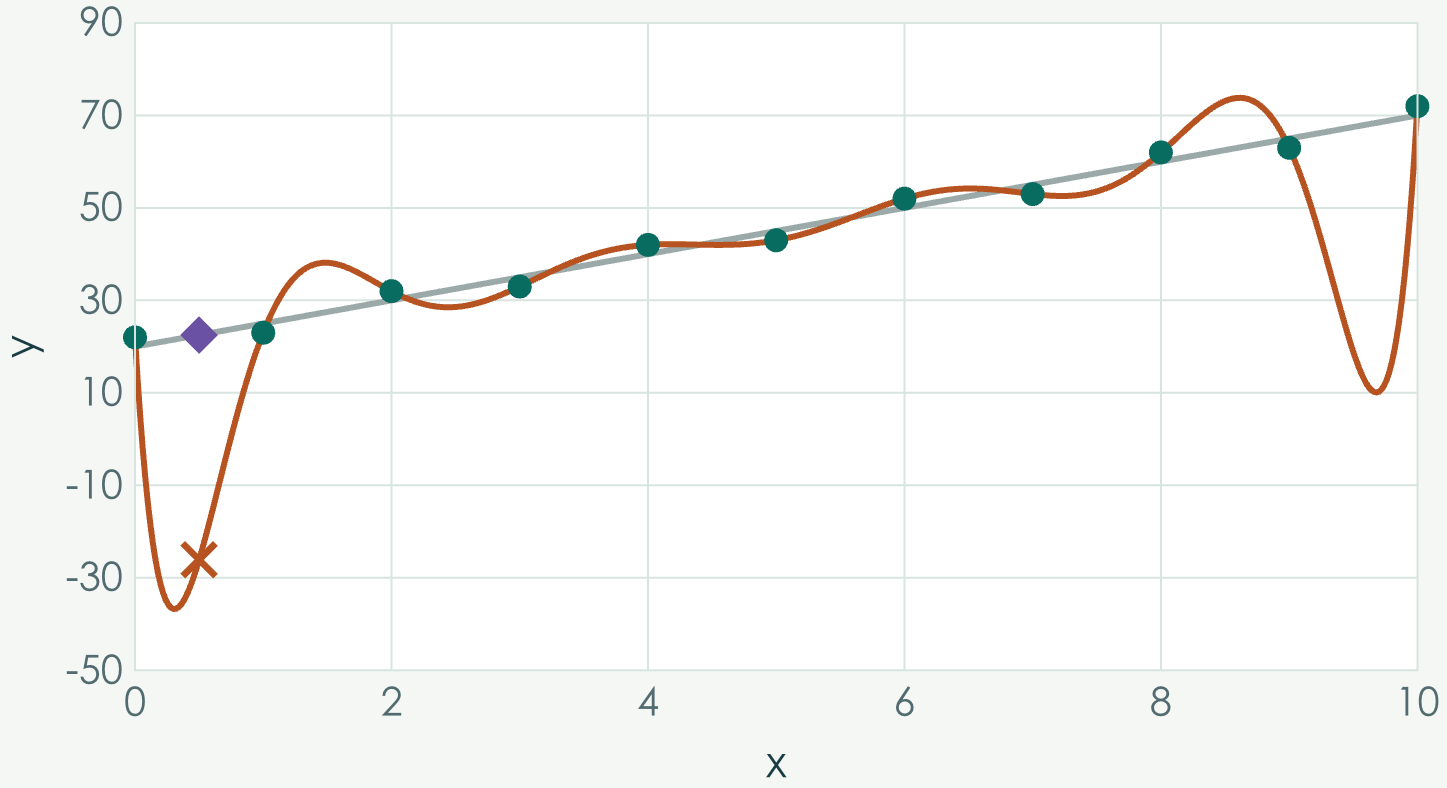
Higher R^2: a better fit to the data used to build the model

用未參與建模的新數據，比較預測與實際結果

Use new data kept out of fitting; compare predictions with actual results

過度擬合：訓練 R^2 很高，仍可猜錯
Overfitting: high training R^2, poor new prediction

綠：訓練　紫：新測試　橙：曲線／預測　灰：原本趨勢



Training R^2 = 1.000

新測試（未用於建模）

New test: not used to fit

x = 0.5

實際 22.5；預測 −26.1

Actual 22.5; predicted −26.1

追著小差異，忽略整體趨勢

Chasing quirks, missing the trend

灰線：模擬數據的原本趨勢 / Grey: underlying simulated trend. Curve model; today we use a line.

帶走你的三個發現
Take away three discoveries

散點讓我們看見關係，也看見例外

Dots show a relationship and its exceptions

R^2 幫助比較直線解釋多少數據差異

R^2 helps compare how much variation a line explains

好擬合不等於好新預測：還要用新數據檢查

Check new predictions against actual results